3. Consider the linear program (primal problem)

$$\begin{aligned} \max \quad & 4x_1 - x_2 + 3x_3 \\ \text{s.t.} \quad & 5x_1 - x_2 + 2x_3 \le 4 \\ & -2x_1 + x_2 + x_3 \le 3 \\ & x_1,\ x_2,\ x_3 \ge 0. \end{aligned}$$

Using complimentary slackness, check whether $\pi = \left(\frac{4}{3}, \frac{1}{3}\right)$ is an optimal solution to the dual of the linear program. Also, find an optimal solution of the primal problem.

primal: $\max\ 4x_1 - x_2 + 3x_3$

$$\begin{aligned} \text{s.t.}\ & 5x_1 - x_2 + 2x_3 \le 4 \\ & -2x_1 + x_2 + x_3 \le 3 \\ & x_1,\ x_2,\ x_3 \ge 0 \end{aligned}$$

Dual: $\min\ 4\pi_1 + 3\pi_2$

$$\begin{aligned} \text{s.t.}\ & 5\pi_1 - 2\pi_2 \ge 4 \\ & -\pi_1 + \pi_2 \ge -1 \\ & 2\pi_1 + \pi_2 \ge 3 \\ & \pi_1,\ \pi_2 \ge 0 \end{aligned}$$

Define $s_i = a_i^T x - b_i \quad (i = 1, 2)$

$s'_j = c_j - \pi^T A_j \quad (j = 1, 2, 3)$

$u_i = \pi_i s_i \ , \ v_j = s'_j x_j$

we have: $u_i = v_j = 0$

$s'_1 = 4 - (5 \times \frac{4}{3} - 2 \times \frac{1}{3}) = -2 \Rightarrow x_1 = 0$

$\because \pi_1, \pi_2 \neq 0 \Rightarrow \begin{cases} s_1 = 0 \\ s_2 = 0 \end{cases} \Rightarrow \begin{cases} -x_2 + 2x_3 - 4 = 0 \\ x_2 + x_3 - 3 = 0 \end{cases}$

$\Rightarrow \begin{cases} x_2 = 2/3 \\ x_3 = 7/3 \end{cases}$ Since $x = (0, \frac{2}{3}, \frac{7}{3})$ is a feasible solution of primal problem,

we know that $\pi = (\frac{4}{3}, \frac{1}{3})$ is the optimal solution of the dual problem.